

Chapter 1

Energy Profiling of 5G User Plane Functions

1.1 Introduction

The User Plane Function (UPF) is the component of a 5G core network responsible for routing and forwarding GTP-U encapsulated user traffic between the Radio Access Network (RAN) and the Data Network (DN). Its power consumption is therefore directly coupled to network load, and understanding this relationship is a prerequisite for energy-aware network management and for meeting increasingly stringent sustainability targets in mobile telecommunications.

This chapter presents a systematic empirical study of two open-source UPF implementations under controlled laboratory conditions. The first implementation, SD-CORE DPDK, is based on the SD-Core project and leverages the Data Plane Development Kit (DPDK) poll-mode I/O driver. The second, USR-UPF, is the user-plane component of OpenAirInterface and uses a conventional interrupt-driven Linux I/O model. Traffic was generated with a Keysight LoadCore emulator, and power was measured at the host level using the Scaphandre process-level power monitor with 1 s sampling resolution.

Place in the thesis. This thesis concerns *augmented observability* for the 5/6G data plane, and this chapter extends the observability envelope along its first axis: it makes observable what a production core cannot natively measure. Hardware energy counters (RAPL) report per-*package* power, not the power drawn by an individual network function; per-UPF attributed energy and QoS compliance are, in an operational system, effectively unmeasurable. The profiling campaign manufactures those signals from controlled measurement, and the fitted surrogate that closes the chapter turns them into a *virtual sensor* the rest of the stack can query. Every downstream component—the forecaster, the twin, and the controllers—ultimately grounds its decisions in the quantities made observable here.

The remainder of this chapter is organised as follows. Section ?? describes the measurement methodology and dataset. Section ?? characterises each variant across performance and energy dimensions. Section ?? investigates the saturation behaviour of USR-UPF in depth. Section ?? analyses the Specific Energy Consumption metric and its interpretation for poll-mode versus interrupt-driven drivers. Section ?? reports feature correlations relevant to predictive energy modelling. Section ?? discusses implications for network operation and modelling. Section ?? presents the two-layer digital twin architecture and its predictive performance. Section ?? describes the NetMob 2023 compatibility adapter.

1.2 Measurement Methodology

1.2.1 Testbed Configuration

Both UPF variants were deployed on identical server hardware. LoadCore injected downlink UDP flows with a fixed packet size of 1250 B at progressively increasing offered loads. Each load step was sustained for a fixed duration to allow steady-state measurements; the raw time-series data were then merged with per-second power readings from Scaphandre.

1.2.2 Dataset

After preprocessing, the merged dataset comprises 9556 one-second measurement samples across 110 USR runs and an equivalent number of DPDK runs. Following feature engineering (removal of zero-variance and near-duplicate columns, unit conversions, target computation), the analysis dataset contains 8781 samples and 44 columns, split evenly between the two variants (4388 DPDK, 4393 USR).

1.2.3 Metrics

The following primary metrics are reported throughout this chapter:

Power (W) Total host power consumption in watts, as reported by Scaphandre (`power_watts`).

Net power (W) Incremental power above the idle baseline: $P_{\text{net}} = P_{\text{total}} - P_{\text{idle}}$.

SEC_{total} Specific Energy Consumption based on total power: $\text{SEC}_{\text{total}} = P_{\text{total}}/\lambda$, where λ is throughput in Gbps.

SEC_{net} Marginal energy cost: $\text{SEC}_{\text{net}} = P_{\text{net}}/\lambda$.

Packet loss Count of GTP-U packets lost per one-second interval at the DN-side receiver.

Downlink delay Weighted mean one-way packet delay on the downlink path (μs).

1.3 Variant Profiles

1.3.1 Overview

Table ?? summarises the key performance and energy characteristics of both variants across all measurement conditions. Figure ?? illustrates the fundamental difference in the power–throughput relationship.

Table 1.1: UPF variant profiles: key performance and energy metrics.

Metric	SD-Core DPDK	USR-UPF
<i>Power</i>		
Total power range (W)	0.807–0.859	0.000–4.497
Mean power (W)	0.821	1.435
Standard deviation (W)	0.005	1.527
Dynamic range (W)	0.052	4.497
Net power mean (W)	0.001	1.419
<i>CPU</i>		
Mean CPU (%)	1.064	1.549
Max CPU (%)	1.079	4.554
Idle CPU (%)	1.064	0.018
<i>Throughput</i>		
Maximum (Gbps)	5.06	2.33
Mean (Gbps)	0.788	0.528
<i>Reliability</i>		
Packet loss (sample rows, %)	0.0	32.7
<i>Latency</i>		
DL delay mean (μ s)	311	5031
DL delay max (μ s)	24978	25000
RAN jitter mean (μ s)	1.3	1428
<i>Energy efficiency at >1 Gbps</i>		
SEC _{total} median (J/Gb)	0.27	1.61
SEC _{net} median (J/Gb)	0.002	1.61
L2/L3 overhead ratio (mean)	2.11	1.86

1.3.2 SD-Core DPDK

The DPDK data-plane driver operates in poll mode: a dedicated CPU thread continuously spins on the receive queue, eliminating interrupt overhead at the cost of occupying the core regardless of traffic. This architectural choice manifests clearly in the measurements.

Power consumption remains almost constant at 0.821 ± 0.005 W across the entire load range (0–5.06 Gbps), yielding a dynamic range of only 52 mW. CPU utilisation is similarly

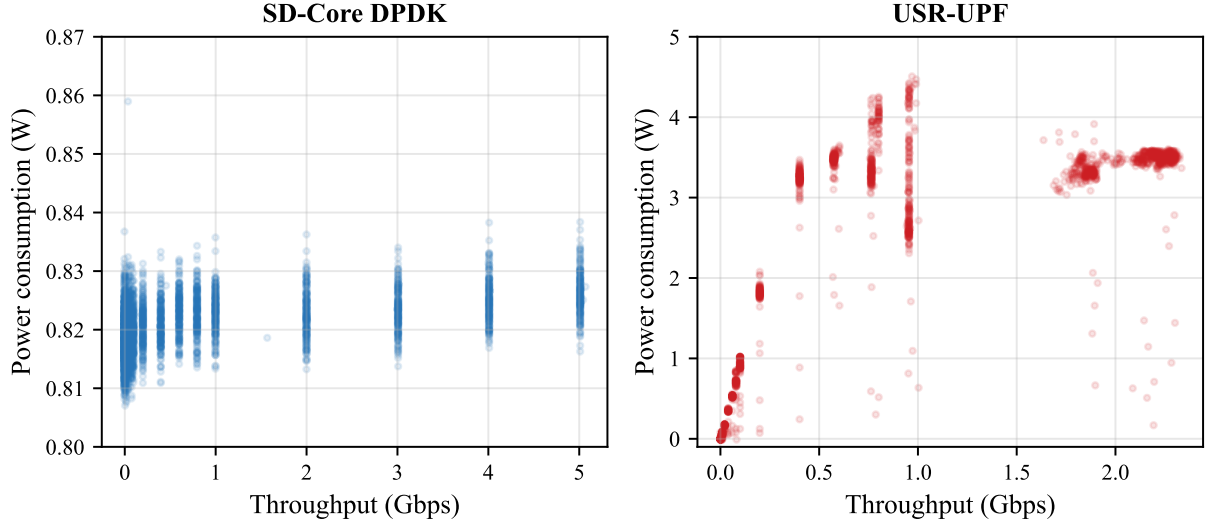


Figure 1.1: Power consumption versus throughput for each UPF variant. SD-Core DPDK (left) exhibits a near-constant power profile due to continuous CPU polling. USR-UPF (right) shows a strong load-proportional relationship but saturates above 0.5 Gbps.

invariant at $1.064 \pm 0.004\%$, and the Pearson correlation between throughput and CPU is $r = 0.016$, confirming that the polling loop consumes a fixed share of the core irrespective of packet rate.

Despite its flat power profile, SD-CORE DPDK achieves the highest throughput of both variants (5.06 Gbps) with zero packet loss across all 4388 measurement samples. Downlink delay is low (mean $311 \mu\text{s}$) and RAN jitter is negligible (mean $1.3 \mu\text{s}$), making it the preferable choice for latency-sensitive traffic.

1.3.3 USR-UPF

USR-UPF uses the standard Linux socket and eBPF/XDP interrupt path. Power scales strongly with offered load: $r(\text{throughput}, \text{power}) = 0.81$ (compared with 0.46 for DPDK), and the system transitions from near-zero power (0.034 W) at idle to a plateau of approximately 3.4 W once saturated. CPU utilisation follows the same trend ($r = 0.76$), rising from 0% at idle to 4.6% at peak.

However, USR-UPF saturates at approximately 0.5 Gbps (see Section ??), beyond which virtually all packets are dropped. Delay is consistently higher than DPDK (mean 5 ms) and RAN jitter degrades catastrophically above saturation (mean $1428 \mu\text{s}$, maximum 199 ms).

Figure ?? illustrates the load-stratified comparison of both variants across the four key metrics.

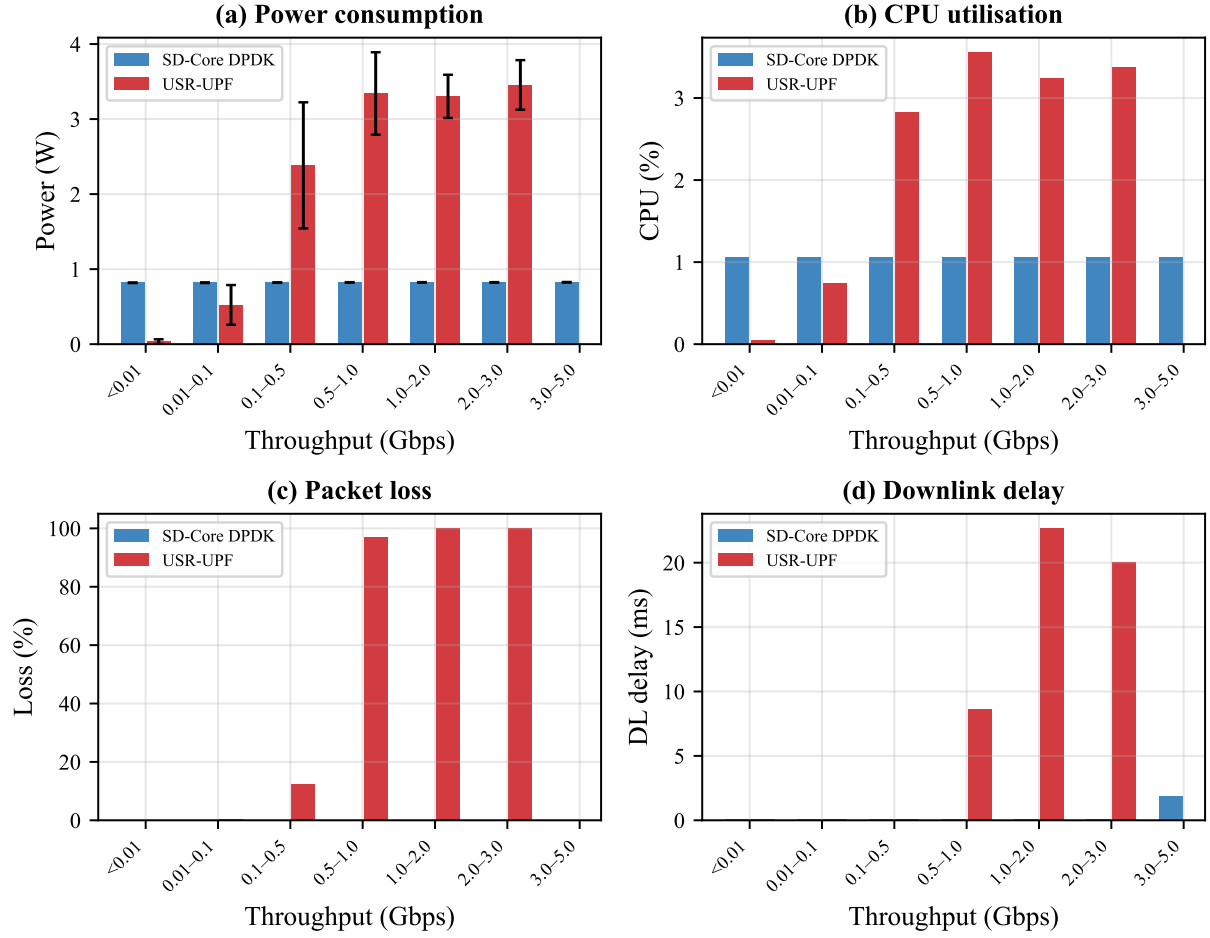


Figure 1.2: Load-stratified variant profiles. Throughput bins on the x-axis (Gbps). (a) Power consumption; (b) CPU utilisation; (c) packet loss incidence; (d) mean downlink delay. Error bars in (a) denote ± 1 standard deviation. DPDK maintains flat power and zero loss at all loads; USR scales with load but saturates above 0.5 Gbps.

1.4 USR-UPF Saturation Analysis

1.4.1 Saturation Onset

To characterise the saturation regime with fine granularity, measurements were binned into 50 Mbps throughput intervals. Figure ?? shows the transition across four dimensions: loss incidence, loss magnitude, delay, and the power/CPU trajectory.

The first bin where more than 5 % of samples exhibit packet loss occurs at ≈ 0.17 Gbps, indicating the onset of saturation. Loss increases steeply, reaching 50 % at ≈ 0.57 Gbps, beyond which essentially all samples report loss. From this point, USR-UPF enters a fully saturated regime: throughput measurements reflect the offered load injected by LoadCore, not the effective forwarding rate, and delay jumps to the measurement ceiling of 25 ms.

USR-UPF Saturation Transition

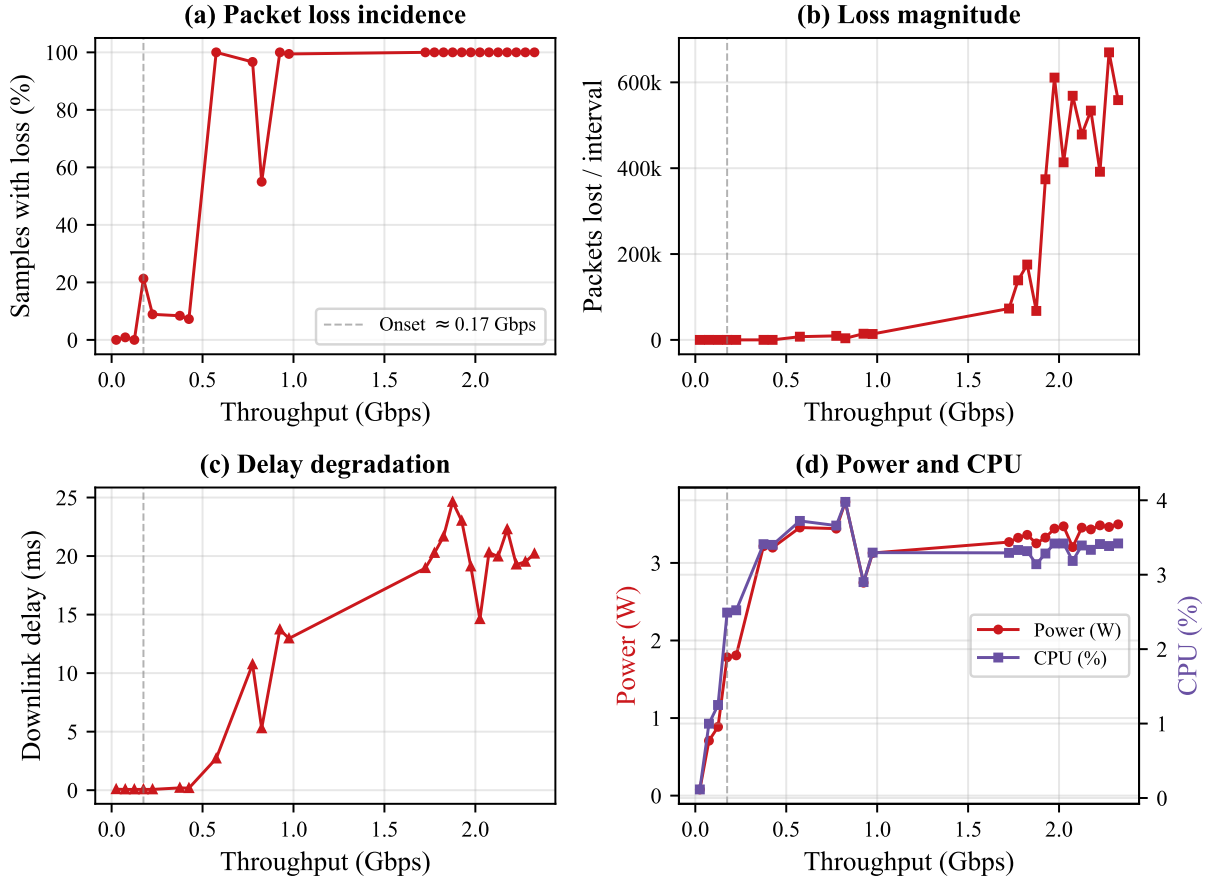


Figure 1.3: USR-UPF saturation transition in 50 Mbps throughput bins. The dashed vertical line marks the saturation onset at ≈ 0.17 Gbps. (a) Fraction of samples with packet loss; (b) mean number of packets lost per interval (when loss occurs); (c) mean downlink delay; (d) mean power and CPU utilisation (dual axis).

1.4.2 Power Behaviour at Saturation

Despite losing packets, USR-UPF continues to consume power proportionally to the offered load up to ≈ 0.5 Gbps, after which power plateaus at approximately 3.4 W (Figure ??d). This indicates that the CPU is fully committed to packet processing at saturation; additional offered load does not increase power but increases loss instead.

The behaviour is summarised as follows:

- Pre-saturation (< 0.3 Gbps): mean power = 0.338 W, zero loss.
- Transition (0.3–0.57 Gbps): power rises sharply, loss begins.
- Post-saturation (> 0.5 Gbps): mean power ≈ 3.4 W, near-total loss.

1.4.3 Per-Run Characterisation

Figure ?? aggregates the loss and power behaviour at the granularity of individual measurement runs. Of the 110 USR runs, 61 (55%) completed without any packet loss; these runs had a maximum offered throughput of at most 0.1 Gbps. Every run exceeding 0.06 Gbps maximum offered load incurred at least some packet loss, establishing the *practical safe operating boundary* of USR-UPF at approximately 100 Mbps.

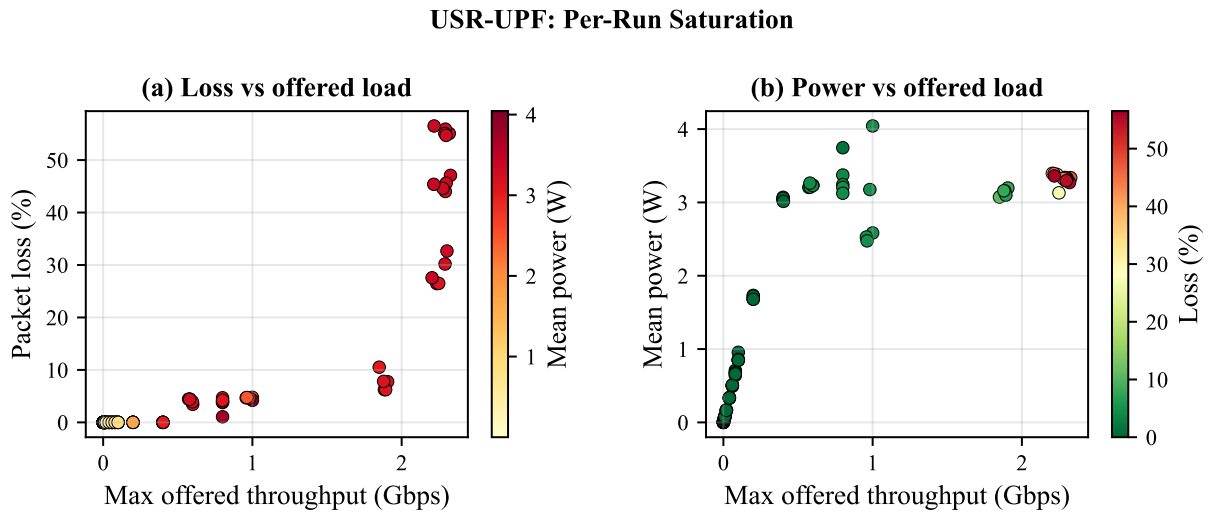


Figure 1.4: USR-UPF per-run saturation summary. Each point represents one measurement run. (a) Overall packet loss versus maximum offered throughput, coloured by mean power; (b) mean power versus maximum offered throughput, coloured by packet loss fraction.

1.5 Specific Energy Consumption Analysis

1.5.1 Definitions

Specific Energy Consumption (SEC) is defined as the ratio of power consumed to useful throughput delivered. Two variants are considered:

$$\text{SEC}_{\text{total}} = \frac{P_{\text{total}}}{\lambda} \quad [J/Gb] \quad (1.1)$$

$$\text{SEC}_{\text{net}} = \frac{P_{\text{net}}}{\lambda} = \frac{P_{\text{total}} - P_{\text{idle}}}{\lambda} \quad [J/Gb] \quad (1.2)$$

1.5.2 The Poll-Mode Measurement Paradox

Equation (??) was introduced in the literature to capture the *marginal* energy cost of transmitting one additional gigabit, isolating the load-dependent component from the fixed idle overhead. While this definition is appropriate for interrupt-driven systems (where $P_{\text{idle}} \ll P_{\text{total}}$), it produces a misleading result for poll-mode drivers.

In SD-CORE DPDK, the CPU polls continuously regardless of load, so $P_{\text{idle}} \approx P_{\text{total}} \approx 0.821 \text{ W}$ at all times. Consequently, $P_{\text{net}} = P_{\text{total}} - P_{\text{idle}} \approx 0$, and SEC_{net} approaches zero as throughput increases, superficially suggesting near-perfect efficiency. In reality, SD-CORE DPDK consumes 0.82 W regardless of whether it forwards 0 or 5 Gbps.

Table ?? and Figure ?? quantify this discrepancy at high throughput ($> 1 \text{ Gbps}$).

Table 1.2: Specific Energy Consumption at high throughput ($> 1 \text{ Gbps}$): median values.

Metric	SD-Core DPDK	USR-UPF
Total power at $>1 \text{ Gbps}$ (W)	0.824	3.478
$\text{SEC}_{\text{total}}$ (J/Gb)	0.274	1.615
SEC_{net} (J/Gb)	0.002	1.607

1.5.3 Implications for Energy Metric Selection

For a fair cross-variant comparison, $\text{SEC}_{\text{total}}$ is the more honest metric: it reflects the full operational cost of sustaining throughput λ . At $> 1 \text{ Gbps}$, DPDK achieves 0.27 J/Gb and USR achieves 1.61 J/Gb —a $6\times$ difference in total power per bit. However, total power consumption must also be considered in absolute terms: at idle, DPDK draws 0.82 W whereas USR draws less than 0.04 W . The preferable variant therefore depends on the traffic load distribution in deployment.

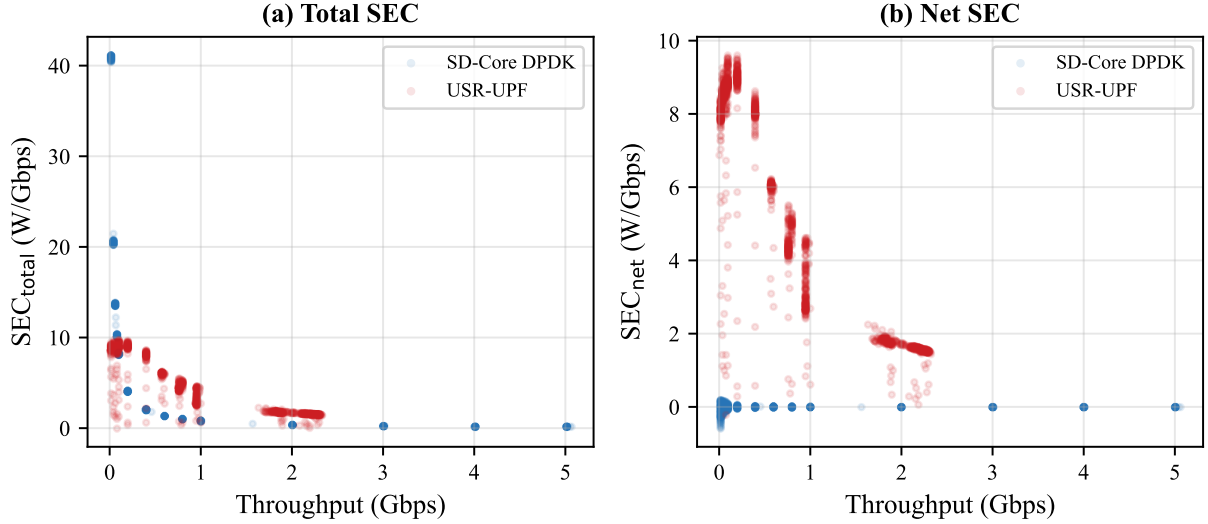


Figure 1.5: Specific Energy Consumption versus throughput. (a) $\text{SEC}_{\text{total}}$: DPDK appears efficient due to low power denominator relative to flat numerator. (b) SEC_{net} : DPDK’s SEC_{net} collapses to near zero because its incremental power $P_{\text{net}} \approx 0$. Both metrics understate the operational energy cost of running a poll-mode UPF. Extreme low-throughput outliers are clipped to the 98th percentile for clarity.

1.6 Feature Correlation Analysis

1.6.1 Correlates of Power Consumption

Table ?? lists the ten strongest Pearson correlations with `power_watts` in the pooled dataset (excluding features derived directly from the target).

Table 1.3: Top feature correlations with power consumption (pooled dataset). Rolling features are marked † to indicate data-leakage risk when used as predictors.

Feature	Pearson r	Note
<code>power_watts_roll_max_30s</code> †	+0.998	Derived from target
<code>cpu_pct</code>	+0.991	Strong legitimate predictor
<code>power_watts_roll_mean_30s</code> †	+0.982	Derived from target
<code>cpu_per_watt</code>	−0.936	Derived from target
<code>dl_delay_power_product</code>	+0.797	Interaction feature
<code>high_delay_frac</code>	+0.768	QoS indicator
<code>weighted_mean_delay_us</code>	+0.735	Latency feature
<code>ul_delay_power_product</code>	+0.614	Interaction feature
<code>l2l3_overhead_ratio</code>	−0.561	Encapsulation efficiency
<code>packets_lost_delta</code>	+0.535	Loss indicator

Figure ?? shows the correlation structure separately for each variant. The two variants exhibit markedly different correlation patterns: in SD-CORE DPDK, power is weakly correlated with all traffic metrics ($r < 0.5$) because the power signal is nearly flat; in USR-UPF, the throughput–power correlation is $r = 0.81$ and CPU–power is $r = 0.92$, making power

highly predictable from traffic observables.

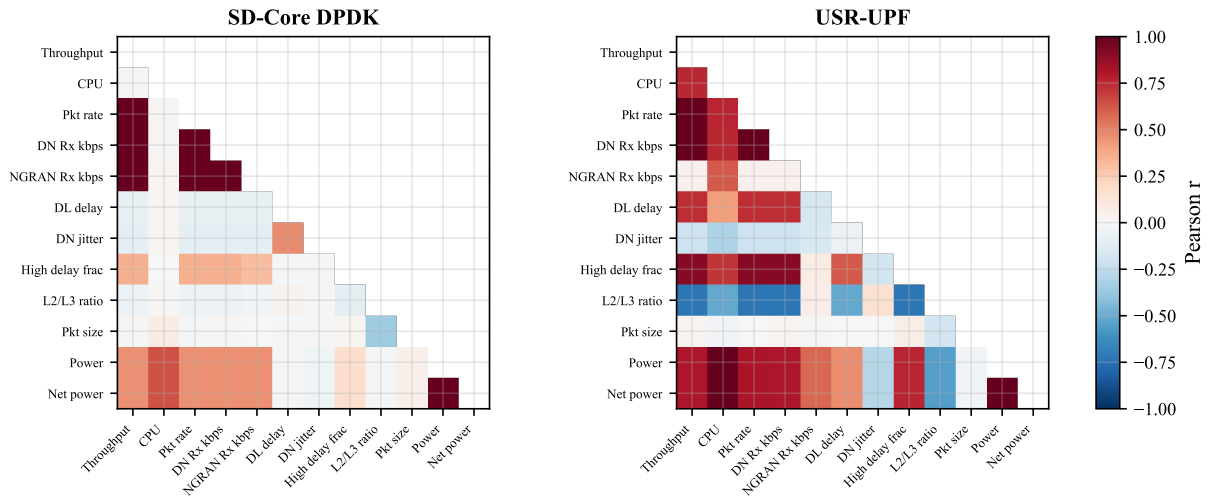


Figure 1.6: Pearson correlation heatmaps for a subset of key features, computed separately for each variant. The DPDK heatmap (left) shows uniformly low correlations with power, consistent with the near-constant power profile. The USR heatmap (right) reveals strong positive correlations between throughput, CPU, and power.

1.6.2 Temporal Dynamics

Figure ?? shows representative power and throughput traces within a single high-load run for each variant. SD-CORE DPDK exhibits essentially no power variation despite large throughput swings; USR-UPF tracks load closely until saturation, then power plateaus while throughput (offered load) continues to increase.

1.7 Discussion and Implications

1.7.1 Architectural Trade-offs

The measurements reveal a fundamental trade-off between the two I/O paradigms:

Poll mode (DPDK) delivers superior throughput (5.06 vs 2.33 Gbps), zero packet loss, and deterministically low latency, at the cost of a fixed power floor of 0.82 W regardless of load. This is appropriate for high-throughput, latency-sensitive deployments where the UPF is expected to be consistently busy.

Interrupt mode (USR) scales power proportionally to load, drawing near-zero power at idle, but saturates at ≈ 0.5 Gbps and incurs severe delay and loss in the saturated regime. It is preferable only in low-to-medium load scenarios where energy-proportionality is prioritised over peak throughput.

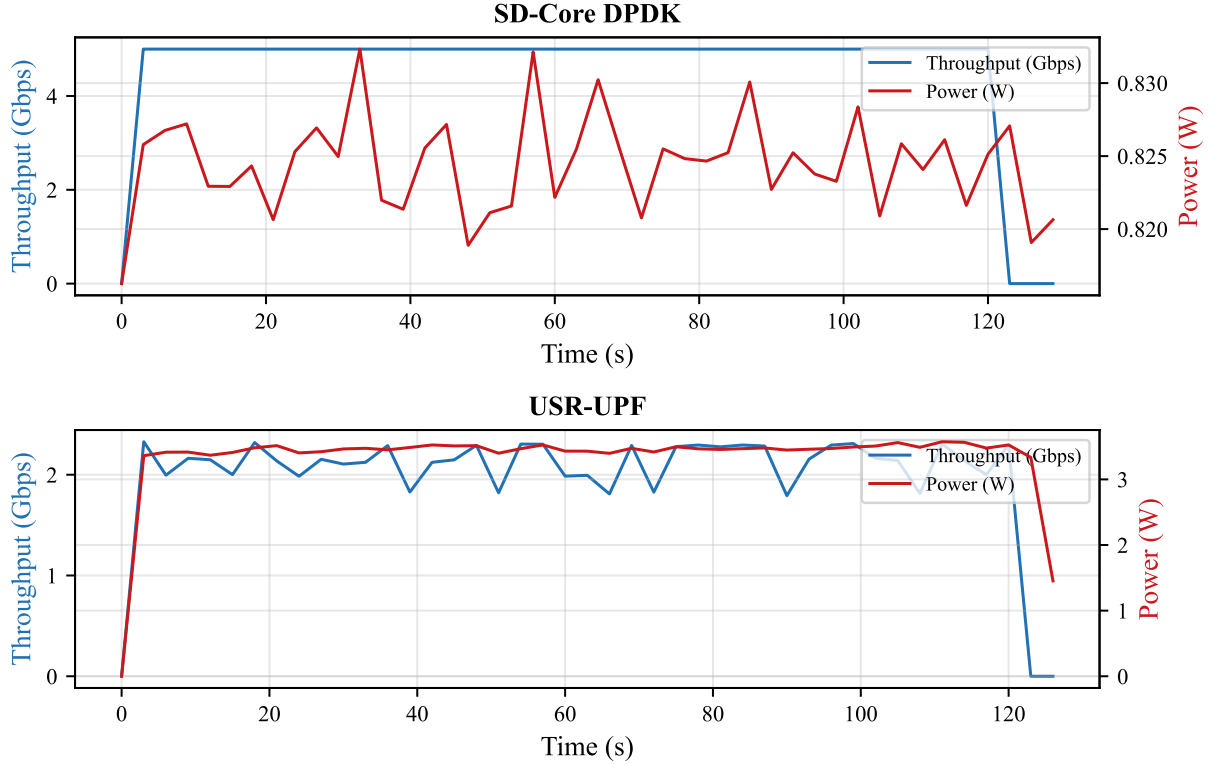


Figure 1.7: Temporal power and throughput traces for representative high-load runs. SD-Core DPDK (top): power remains flat despite throughput variations. USR-UPF (bottom): power rises with load until the saturation plateau.

1.7.2 Safe Operating Regions

Based on the per-run saturation analysis (Section ??, Figure ??), the practical safe operating boundary for USR-UPF is $\lambda \leq 0.10 \text{ Gbps}$. Above this threshold, packet loss becomes probable; above 0.57 Gbps, it is near-certain. No such constraint applies to SD-CORE DPDK within the measurement range.

1.7.3 Implications for Predictive Energy Modelling

The near-constant power of SD-CORE DPDK presents a challenge for regression-based energy models: the target variable has a dynamic range of only 52 mW, approaching the noise floor of Scaphandre. Models trained on DPDK data are expected to predict 0.82 W for almost any input; the information content of the traffic features is minimal. Per-variant models are therefore strongly recommended over a single pooled model.

For USR-UPF, the strong linear relationships ($r > 0.8$) between throughput, CPU, and power suggest that simple linear or low-complexity tree models may suffice in the unsaturated regime. Training should be restricted to the safe operating region ($\lambda < 0.5 \text{ Gbps}$) unless the modelling objective explicitly includes the saturated regime.

The rolling power features (`power_watts_roll_mean_30s`, `power_watts_roll_max_30s`) achieve very high correlations with the power target ($r > 0.98$) and must be excluded as

predictors when the modelling goal is to predict power from non-power observables.

1.8 Variant Performance Summary

Before constructing the digital twin, it is instructive to compare the three deployment variants—DPDK, USR (full range), and USR (safe regime)—across all five key performance indicators (KPIs): downlink throughput, CPU utilisation, packet loss, one-way delay, and power consumption. Figure ?? shows the empirical distributions of each KPI as violin plots, with individual measurements overlaid as semi-transparent scatter points.

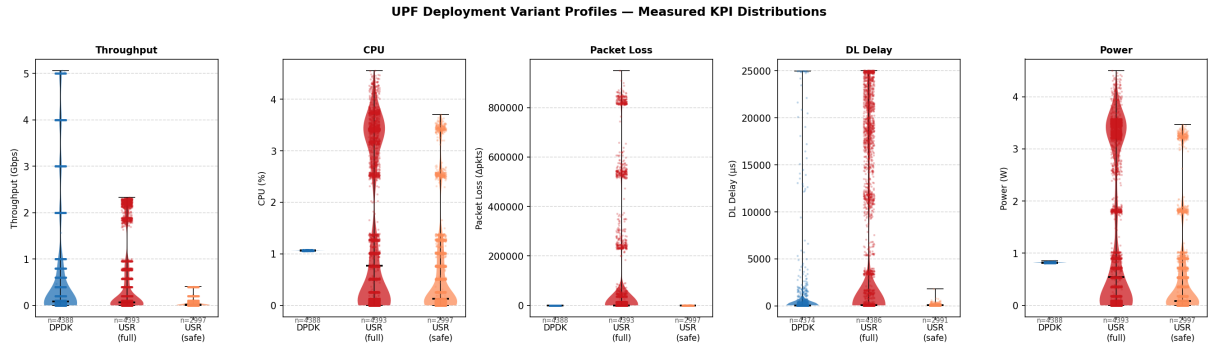


Figure 1.8: Measured KPI distributions for each deployment variant. Each violin combines a kernel-density estimate with the observed range; horizontal bars mark the median. Points are jittered for visibility. DPDK power exhibits near-constant behaviour (poll-mode driver draws fixed power regardless of load), explaining its low model R^2 . USR (safe) packet loss and delay are compressed near zero (traffic kept below the saturation knee at 0.5 Gbps), which limits their model predictability compared with the full-range USR variant.

Three structural differences stand out:

1. **DPDK power is load-independent.** The poll-mode driver keeps all cores spinning at full speed irrespective of traffic; the power distribution is narrow and flat across all load steps. This is the primary reason the DPDK Layer-2 power model achieves only $R^2 = 0.29$.
2. **USR (safe) compresses loss and delay to near-zero.** By restricting the training set to throughput below 0.5 Gbps—the stable linear regime—virtually no packet loss or queuing delay is induced. The resulting low variance makes these two targets harder to fit ($R^2 = 0.86$ and 0.52 respectively) despite the excellent power and CPU models ($R^2 > 0.999$).
3. **USR (full range) covers saturation.** Including the high-load regime exposes the non-linear onset of loss and delay, yielding $R^2 = 0.998$ and 0.968 for those targets at the cost of a slightly noisier power model ($R^2 = 0.988$).

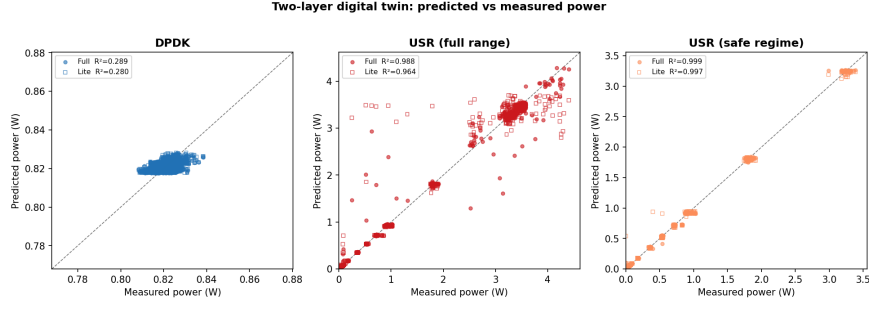


Figure 1.9: Predicted versus measured power for the three trained variants (DDPK, USR full range, USR safe regime). Solid markers: full model; open markers: lite (NetMob-compatible) model. The dashed line is the identity.

The choice between USR safe and USR full therefore depends on the inference context: USR safe is recommended when input loads are expected to remain below the saturation knee (e.g. NetMob 2023 real-world traffic), while USR full is preferable when the full operating envelope—including congestion—must be captured.

1.9 Data-Driven Digital Twin

1.9.1 Architecture

The empirical findings of the preceding sections motivate a *predictive digital twin*: a machine-learning model that, given only an offered traffic load, outputs the UPF’s expected power consumption, CPU utilisation, delivered throughput, packet loss, and one-way delay—without requiring a live UPF deployment.

The twin is organised in two layers that reflect the causal structure of the system (Figure ??).

Layer 1 (performance models) Four independent regression models predict UPF *responses* from the *offered load*: delivered throughput (Gbps), CPU utilisation (%), downlink packet loss (packets per interval), and downlink one-way delay (μ s). The input feature set consists exclusively of TX-side, controlled quantities—DL/UL kbits/s, TX packet counts, average packet size, and the L2/L3 overhead ratio.

Layer 2 (power model) A single model predicts `power_watts` from the same offered-load features *plus* the Layer 1 outputs. Using Layer 1 outputs as Layer 2 features captures the true causal pathway (load $\rightarrow$ CPU $\rightarrow$ power) without introducing circular dependency at inference time, because Layer 1 predictions are available before Layer 2 runs.

1.9.2 Training Strategy

Three variant-specific models are trained to reflect the distinct operating regimes identified in Section ??:

- **dpgk** — all 4 388 DPDK samples. Included as an evidence baseline even though the near-constant power (52 mW range) makes the regression R^2 inherently low.
- **usr_full** — all 4 393 USR samples, including the saturated regime above 0.57 Gbps.
- **usr_safe** — USR samples with delivered throughput below 0.5 Gbps, corresponding to the unsaturated, practically deployable regime.

For each variant and each prediction slot (Layer 1 target or Layer 2 target), three model types are compared: Ridge regression, Random Forest, and Gradient Boosted Trees. All hyperparameter selection uses 5-fold cross-validation on the training split; the model with the highest test-set R^2 is retained. Layer 2 trains on *out-of-fold* Layer 1 predictions produced by a second 5-fold pass, so it never observes ground-truth Layer 1 values during training—matching the distribution it will encounter at inference time (stacking).

1.9.3 Model Performance

Table ?? summarises Layer 2 power model performance. Figure ?? shows predicted versus measured power for all variants and both model variants.

Table 1.4: Layer 2 power model test-set metrics (best model type per slot).

Variant	Mode	R^2	RMSE (W)	MAE (W)	Best model
DPDK	Full	0.289	0.004	0.003	Ridge
DPDK	Lite	—	—	—	—
USR (full)	Full	0.988	0.172	0.050	Random Forest
USR (full)	Lite	0.964	0.245	0.070	Random Forest
USR (safe)	Full	0.999	0.026	0.012	Ridge
USR (safe)	Lite	0.997	0.038	0.017	Ridge

The **usr_safe** variant achieves $R^2 = 0.999$ with the full feature set and $R^2 = 0.997$ with only DL/UL throughput inputs, confirming that power scales nearly linearly with load in the unsaturated regime. The **usr_full** model degrades gracefully ($\Delta R^2 \approx 0.024$) in lite mode, demonstrating that packet-count and overhead features add moderate but non-negligible predictive value. The DPDK $R^2 = 0.289$ confirms the analytical expectation: a 52 mW signal range is indistinguishable from noise at the model’s operating scale.

Figure ?? provides the full R^2 picture across all prediction slots. Figure ?? shows the three-way algorithm comparison for the L2 power model, directly justifying the winning

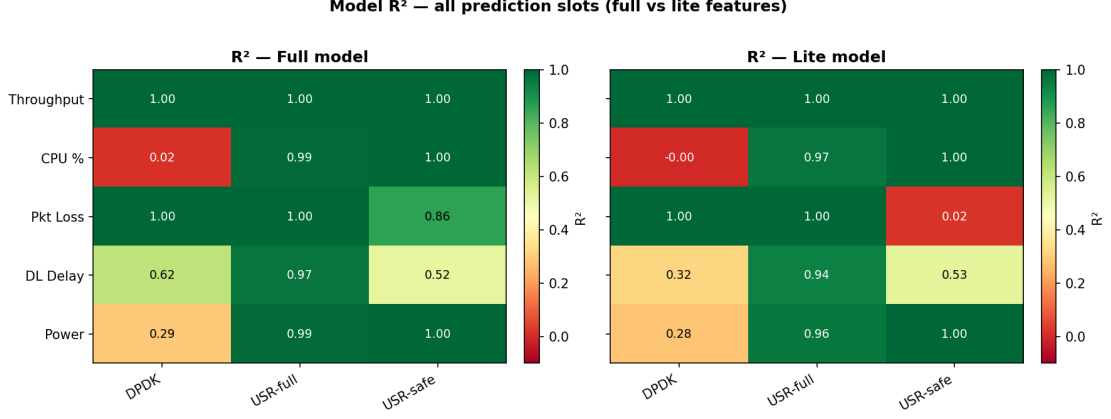


Figure 1.10: R² for all prediction slots (variant × target), full model (left) and lite model (right). Green = high accuracy; red = low. DPDK CPU and DPDK delay are structurally unpredictable (poll-mode artefacts).

selection per variant. Figure ?? extends this comparison to all L1 targets. Figure ?? presents side-by-side interpretability plots (Ridge coefficients and tree importances) on the same feature order, making the two model families comparable regardless of which was selected.

1.10 NetMob 2023 Compatibility

The NetMob 2023 dataset provides per-cell normalised UL/DL traffic loads $\hat{d} \in [0, 1]$ but no packet-level detail. To drive the digital twin from such data a thin adapter layer translates normalised loads into the model’s input space:

$$B_{\text{DL}} = \hat{d}_{\text{DL}} \cdot C_{\text{max,DL}} \cdot 10^6 \quad [\text{kbits/s}] \quad (1.3)$$

$$N_{\text{pkt}} = \frac{B_{\text{DL}} \cdot 10^3 \cdot \Delta t}{8 \cdot \bar{s}_{\text{pkt}}} \quad (1.4)$$

where C_{max} is the rated UPF link capacity (default 10 Gbps), $\Delta t = 3$ s is the sampling interval, and $\bar{s}_{\text{pkt}}$ is the assumed average packet size (default 1250 B). Missing features (overhead ratio, etc.) are imputed from deployment constants stored in `params.yaml`.

Full versus lite models. The *full* model accepts all seven offered-load features and serves laboratory or high-fidelity simulation scenarios. The *lite* companion model accepts only B_{DL} and B_{UL} , making it directly compatible with NetMob and any other dataset where only load volume is known. The lite model quantifies how much accuracy is sacrificed for this interoperability: for `usr_safe`, the power-model R² drops by only 0.002 (from 0.999 to 0.997).

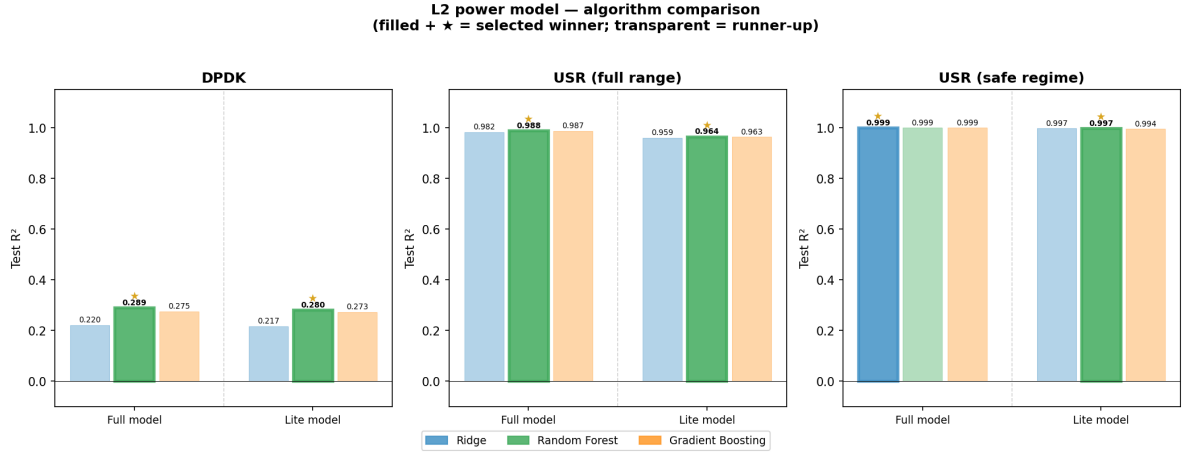


Figure 1.11: Algorithm selection for the L2 power model. Each group shows Ridge, Random Forest, and Gradient Boosting test R^2 for the full and lite feature sets. Filled bars and gold stars (★) mark the selected winner per group; transparent bars are runners-up. Ridge wins for `usr_safe` (linear regime); Random Forest wins for `usr_full` (non-linear saturation effects).

1.11 Summary

This chapter characterised two 5G UPF implementations under controlled load conditions. The principal findings are:

1. **DPDK is power-constant:** total power varies by only 52mW over the full 0–5 Gbps load range due to the CPU polling loop. This makes power prediction from traffic features statistically intractable.
2. **USR is power-proportional but capacity-limited:** power scales linearly from ≈ 0 to ≈ 3.4 W, but packet loss becomes near-universal above 0.57 Gbps. The practical operating boundary is ≈ 0.1 Gbps.
3. **The SEC measurement paradox:** the SEC_{net} metric misleadingly reports near-zero energy efficiency for DPDK because the idle power is almost equal to the load power. Total SEC ($\text{SEC}_{\text{total}}$) is the more operationally meaningful metric.
4. **CPU is the strongest legitimate predictor of power** ($r = 0.99$ pooled), followed by delay-related features. Rolling power features are informative but constitute data leakage when used as predictors of power.
5. **Separate per-variant models are necessary:** the two variants occupy distinct regimes with different predictive structures and cannot be modelled jointly without a variant indicator.
6. **A two-layer digital twin achieves high accuracy for USR:** the `usr_safe`

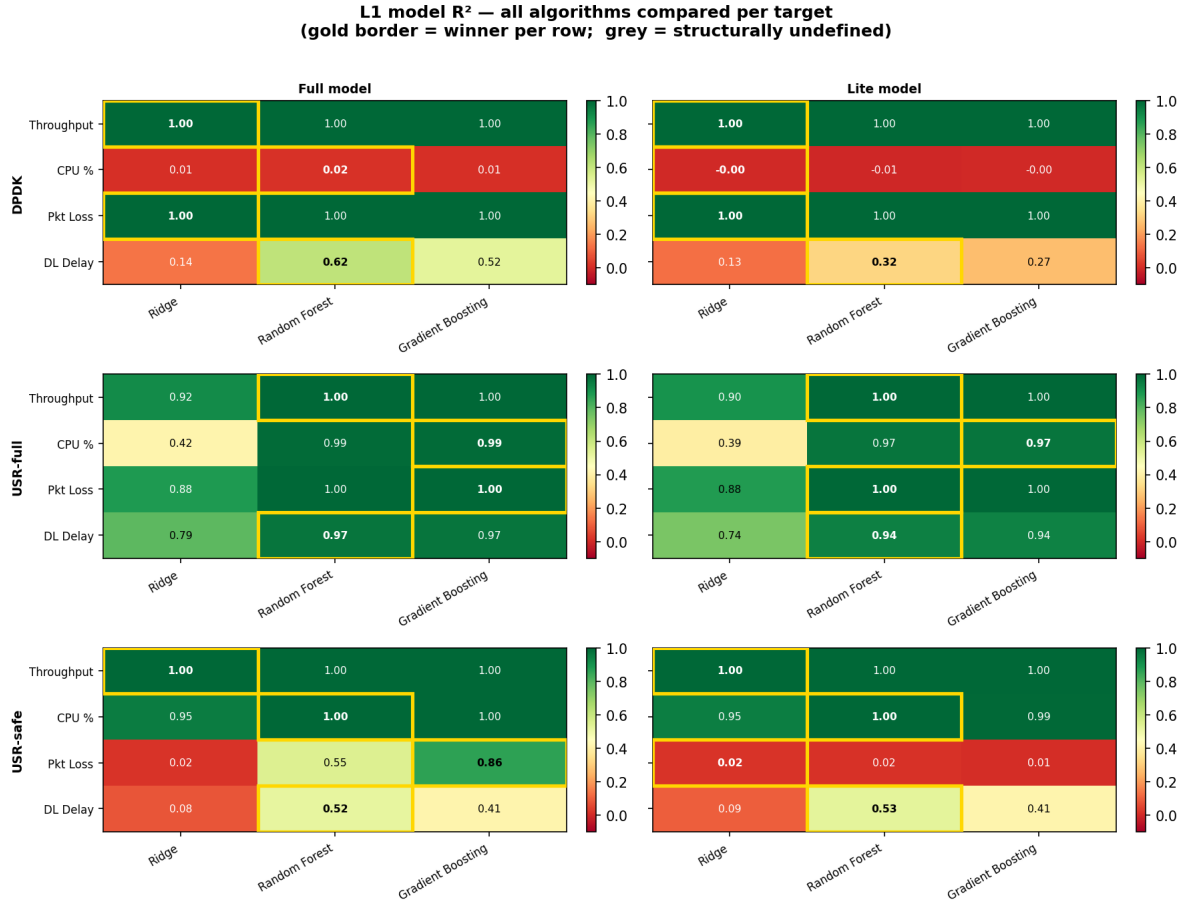


Figure 1.12: Algorithm comparison for all L1 prediction slots. Rows: prediction targets; columns: algorithm types. Gold borders mark the winning algorithm per target row. Ridge dominates the linear throughput–power relationship; tree models offer marginal gains only for CPU and delay targets.

power model reaches $R^2 = 0.999$ (full) and $R^2 = 0.997$ (lite), enabling power prediction from offered load alone without a live UPF.

- NetMob compatibility incurs minimal cost:** the lite model, driven by normalised UL/DL loads only, sacrifices $\Delta R^2 < 0.003$ for `usr_safe`, making the twin directly usable with real-world traffic traces.

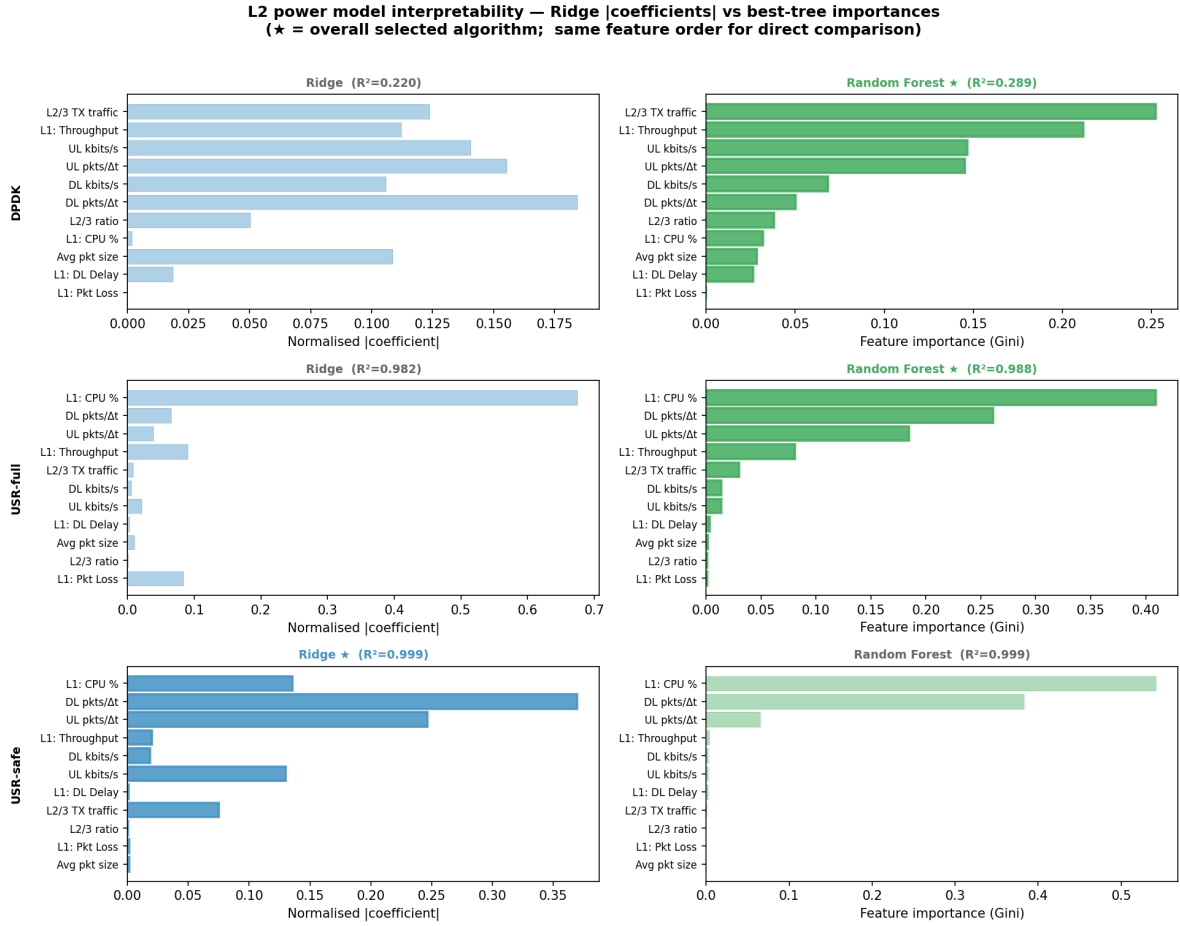


Figure 1.13: L2 power model interpretability for each variant. *Left panels*: Ridge regression normalised absolute coefficients. *Right panels*: feature importances from the best tree-based model (Random Forest or Gradient Boosting). Both panels share the same feature ordering (sorted by tree importance) to allow direct comparison regardless of which algorithm was selected (marked ★). The dominance of L1-predicted CPU and throughput confirms the causal pathway load → CPU → power.